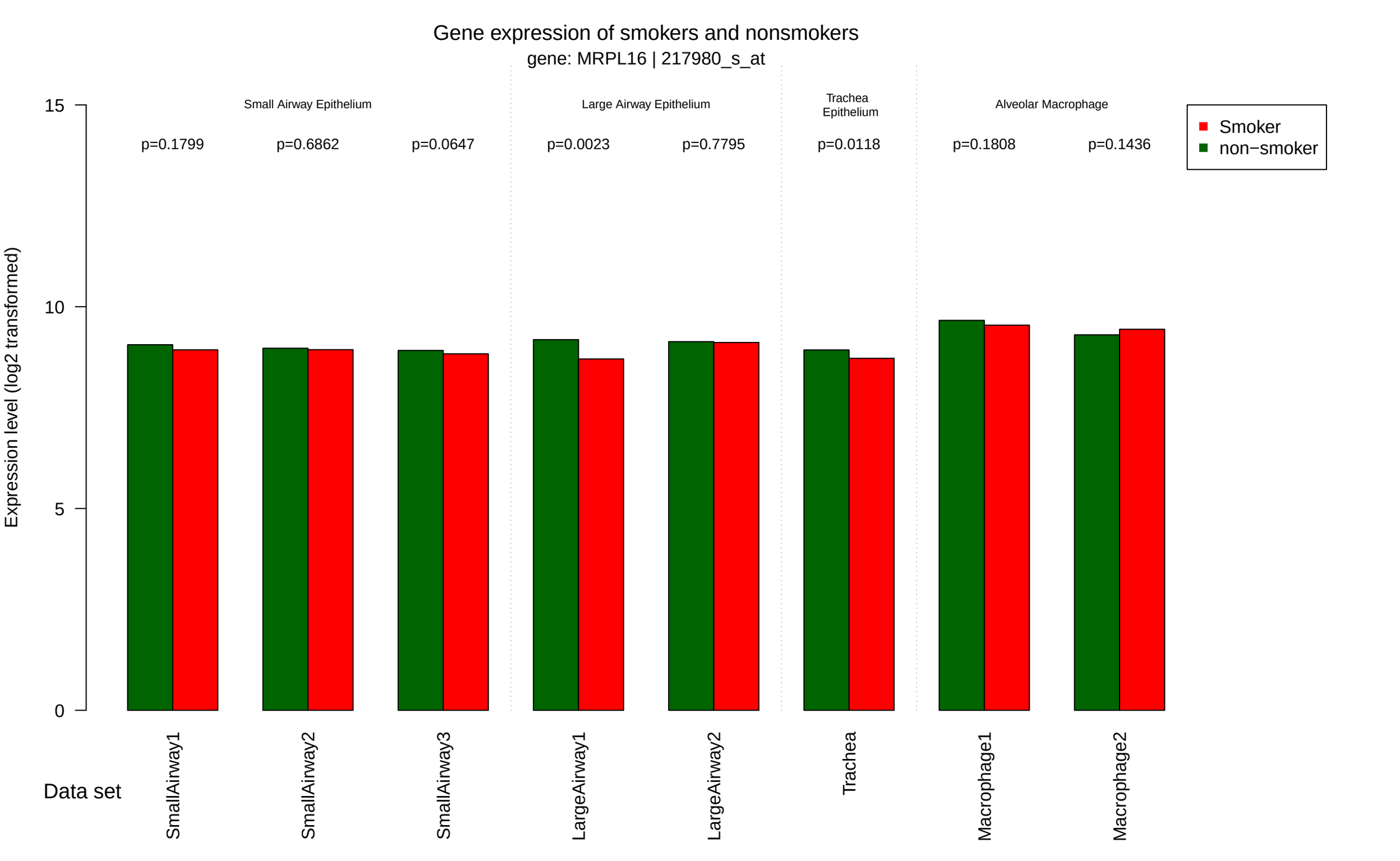
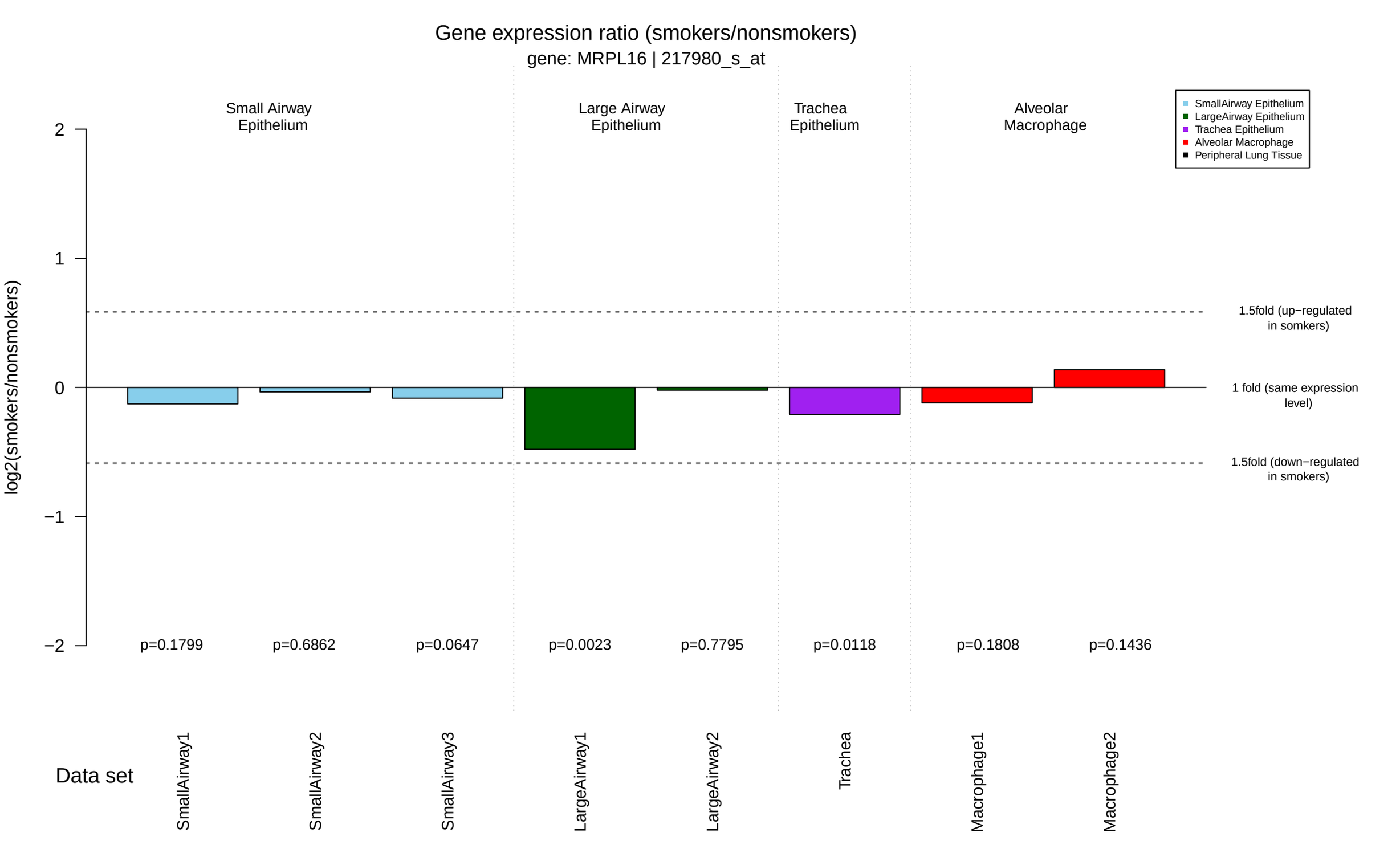
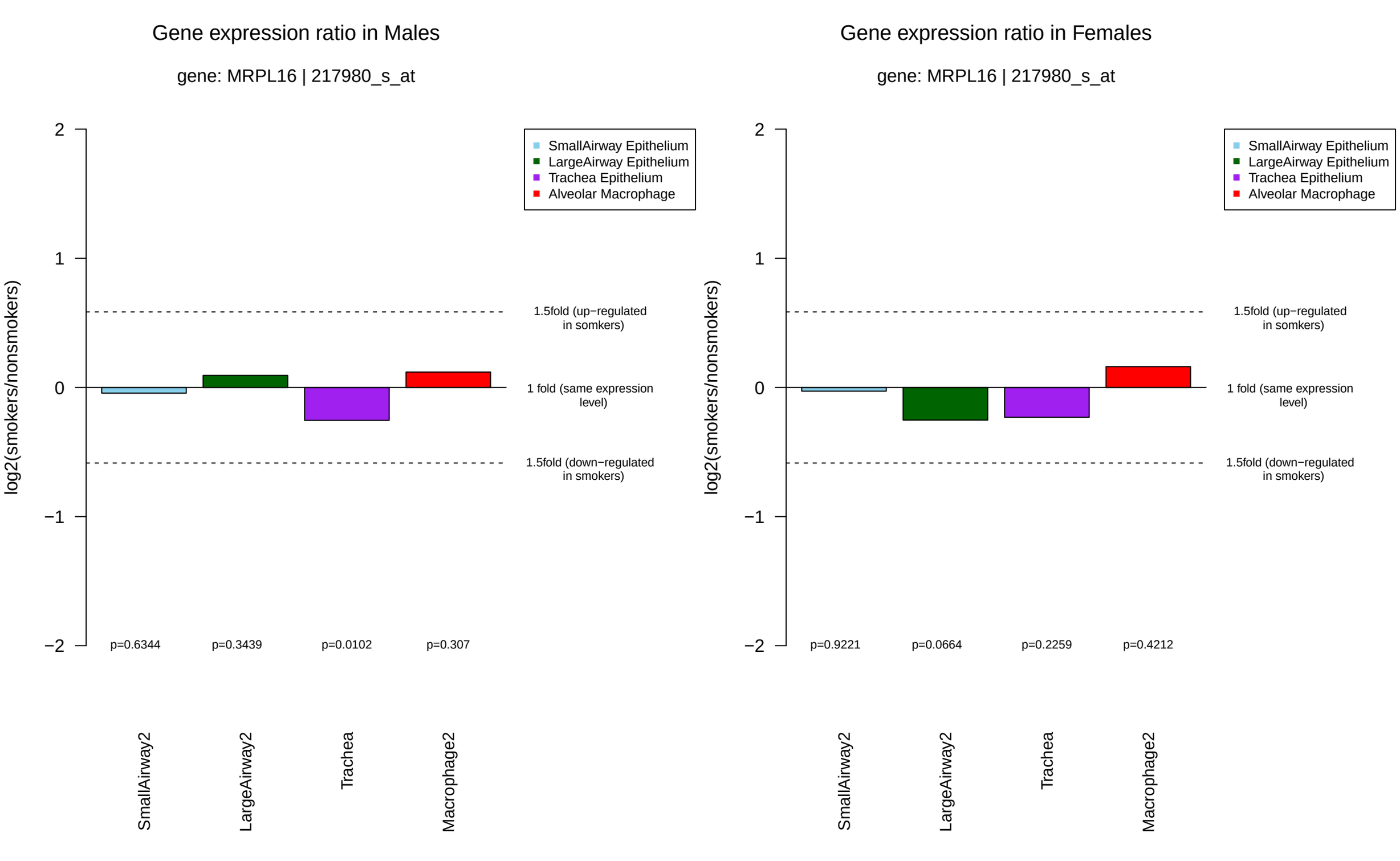
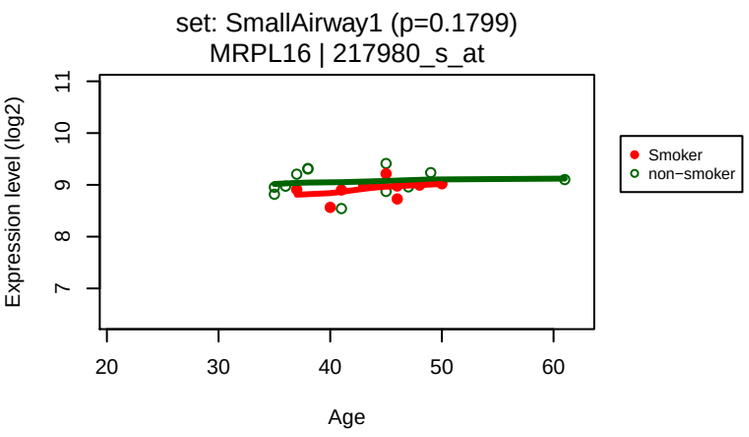
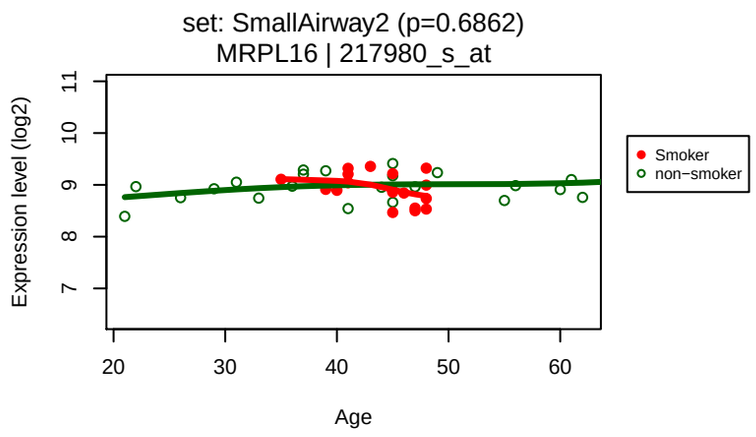
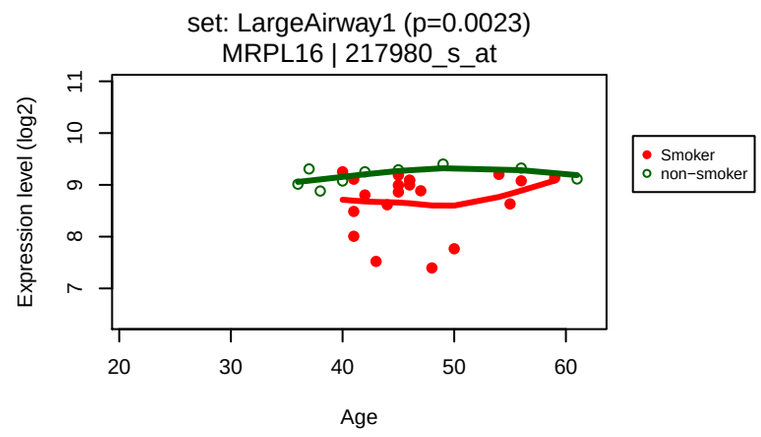

Gene expression of smokers and nonsmokers
gene: MRPL16 | 217980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1799
p=0.6862
p=0.0647
p=0.0023
p=0.7795
p=0.0118
p=0.1808
p=0.1436
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

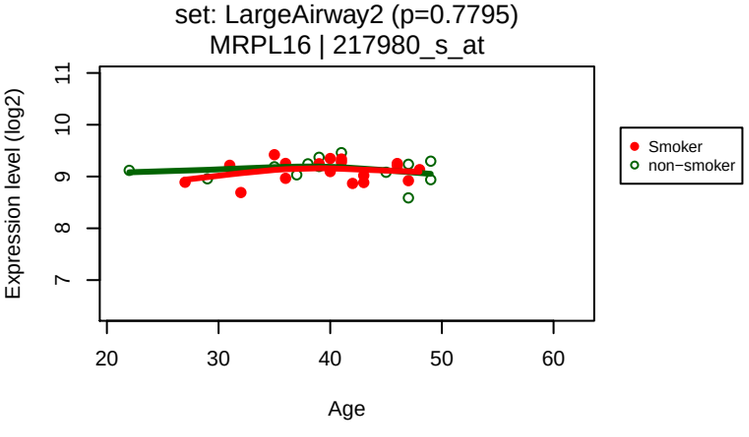

Gene expression ratio (smokers/nonsmokers)
gene: MRPL16 | 217980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1799
p=0.6862
p=0.0647
p=0.0023
p=0.7795
p=0.0118
p=0.1808
p=0.1436
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

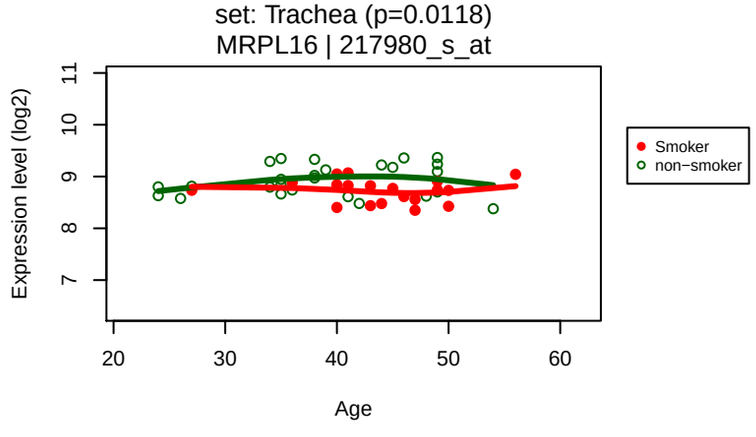

Gene expression ratio in Males
gene: MRPL16 | 217980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6344
p=0.3439
p=0.0102
p=0.307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL16 | 217980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9221
p=0.0664
p=0.2259
p=0.4212
SmallAirway2
LargeAirway2
Trachea
Macrophage2

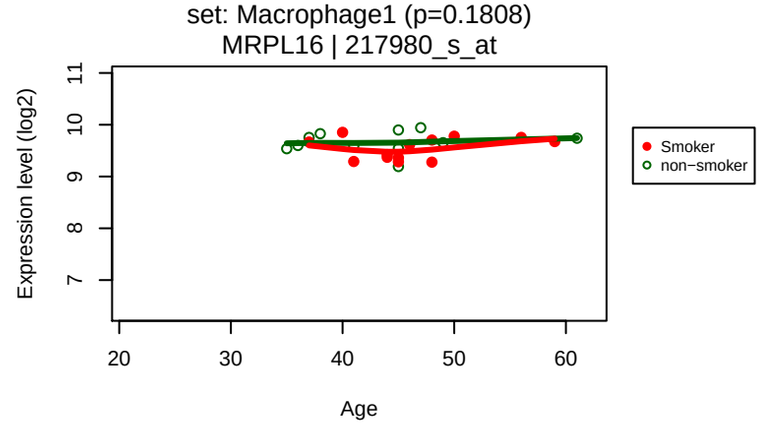

set: SmallAirway1 (p=0.1799)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker

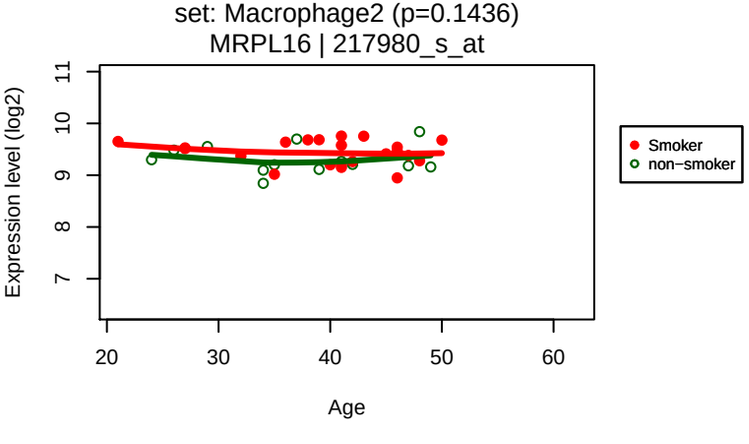

set: SmallAirway2 (p=0.6862)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker

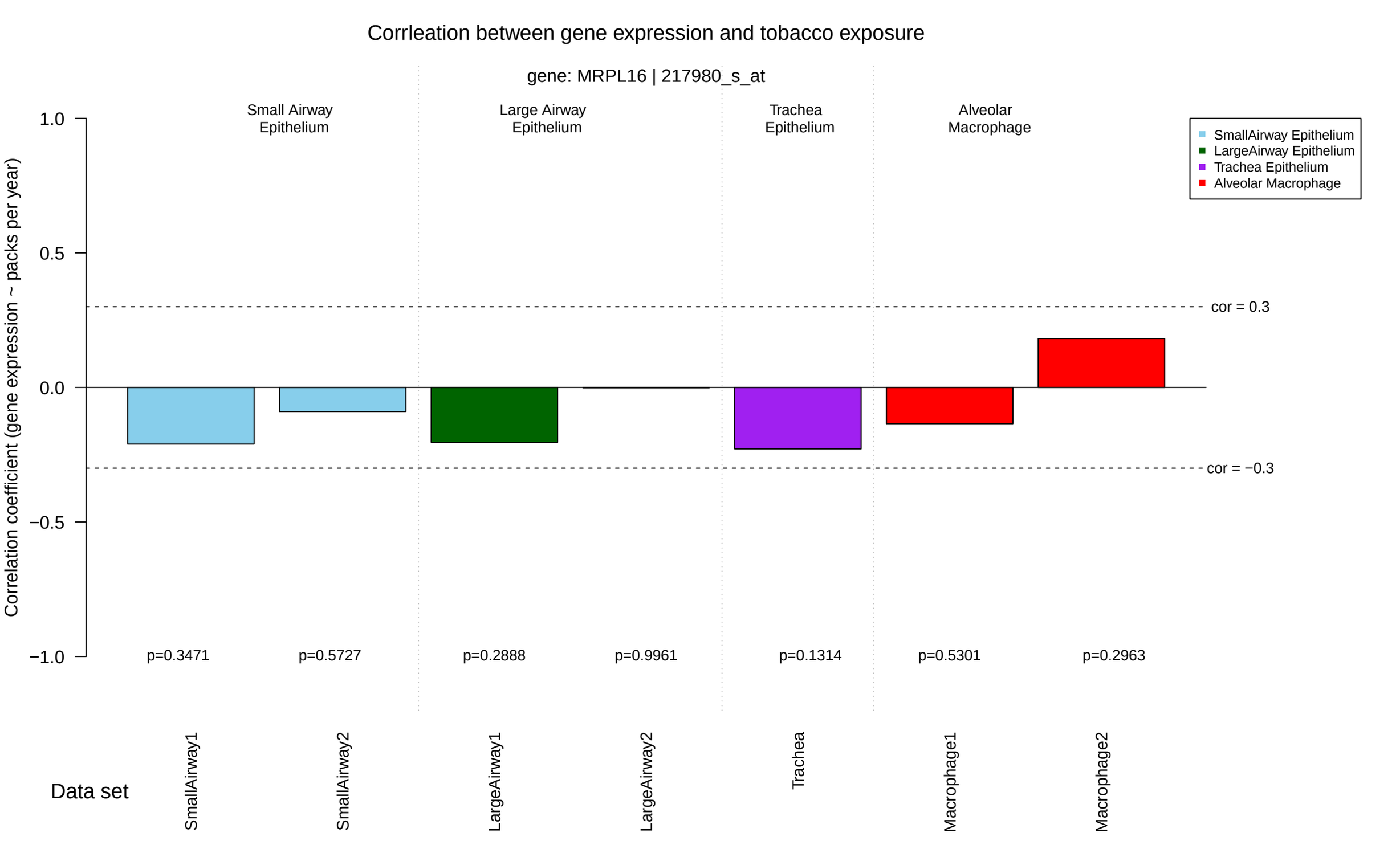

set: LargeAirway1 (p=0.0023)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7795)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0118)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1808)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1436)
MRPL16 | 217980_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL16 | 217980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3471
p=0.5727
p=0.2888
p=0.9961
p=0.1314
p=0.5301
p=0.2963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2